NEXUS AI

AI-Powered Smart Civic Issue Reporting & Governance Platform

Report. Detect. Resolve.
Transform Communities.



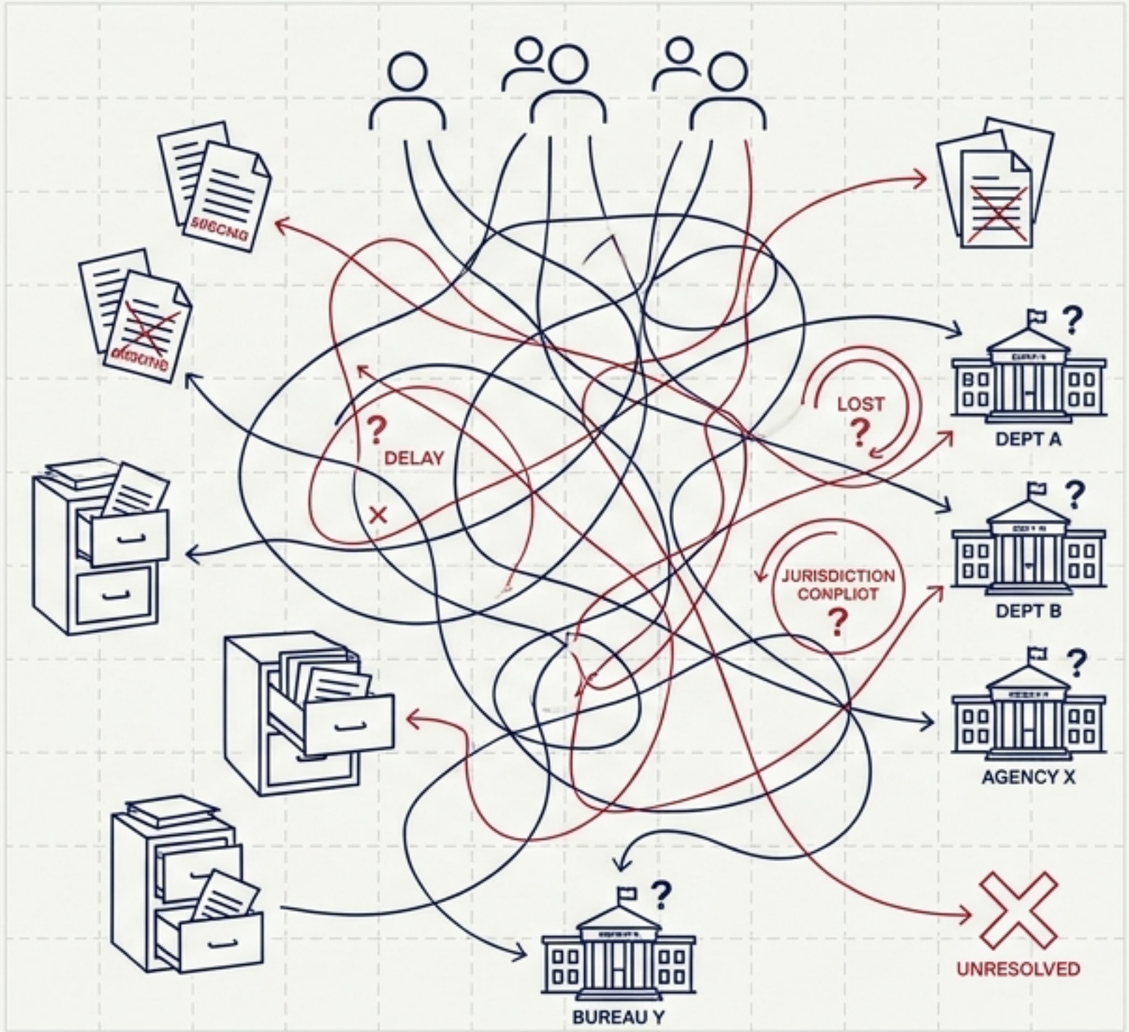
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The Civic Infrastructure Gap

Physical Reality: Real-time Infrastructure Decay

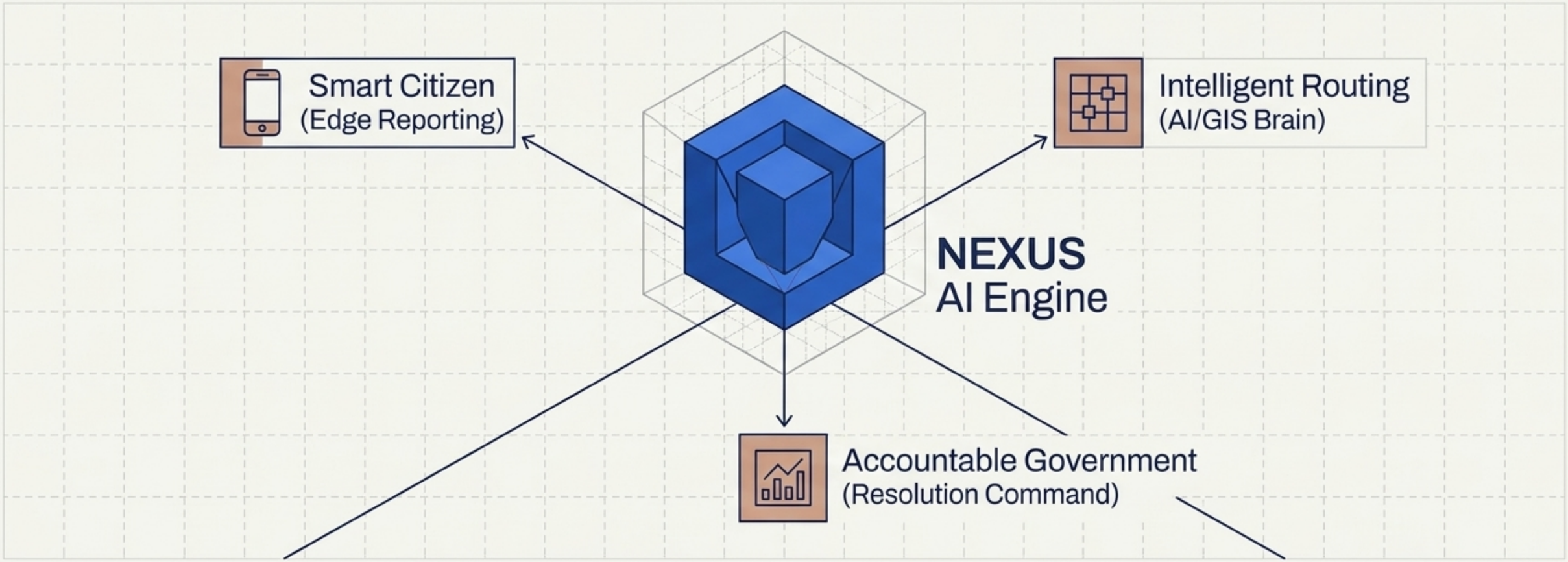


System Reality: Manual processing, jurisdiction confusion, lost paperwork, and zero accountability



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The NEXUS AI Ecosystem



Traditional:	Manual Input	→	Guesswork Routing	→ Trust-Based Closure
NEXUS AI:	Instant Mobile Reporting	→	AI/GIS Auto-Routing	→ AI Visual Verification

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Smart Reporting at the Edge



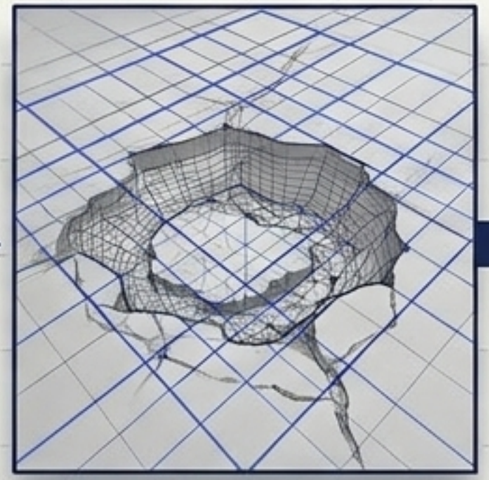
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AI-Powered Civic Issue Detection Pipeline

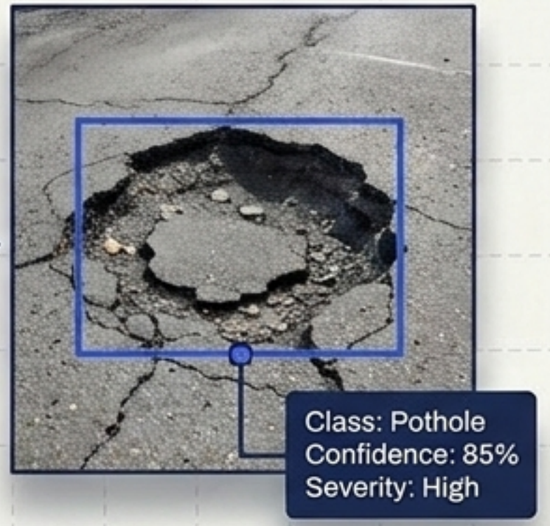
Input



Processing



Output



- Automated Classification
- Severity & Priority Assessment
- Duplicate Complaint Detection (Flags visually similar reports within same GPS radius)

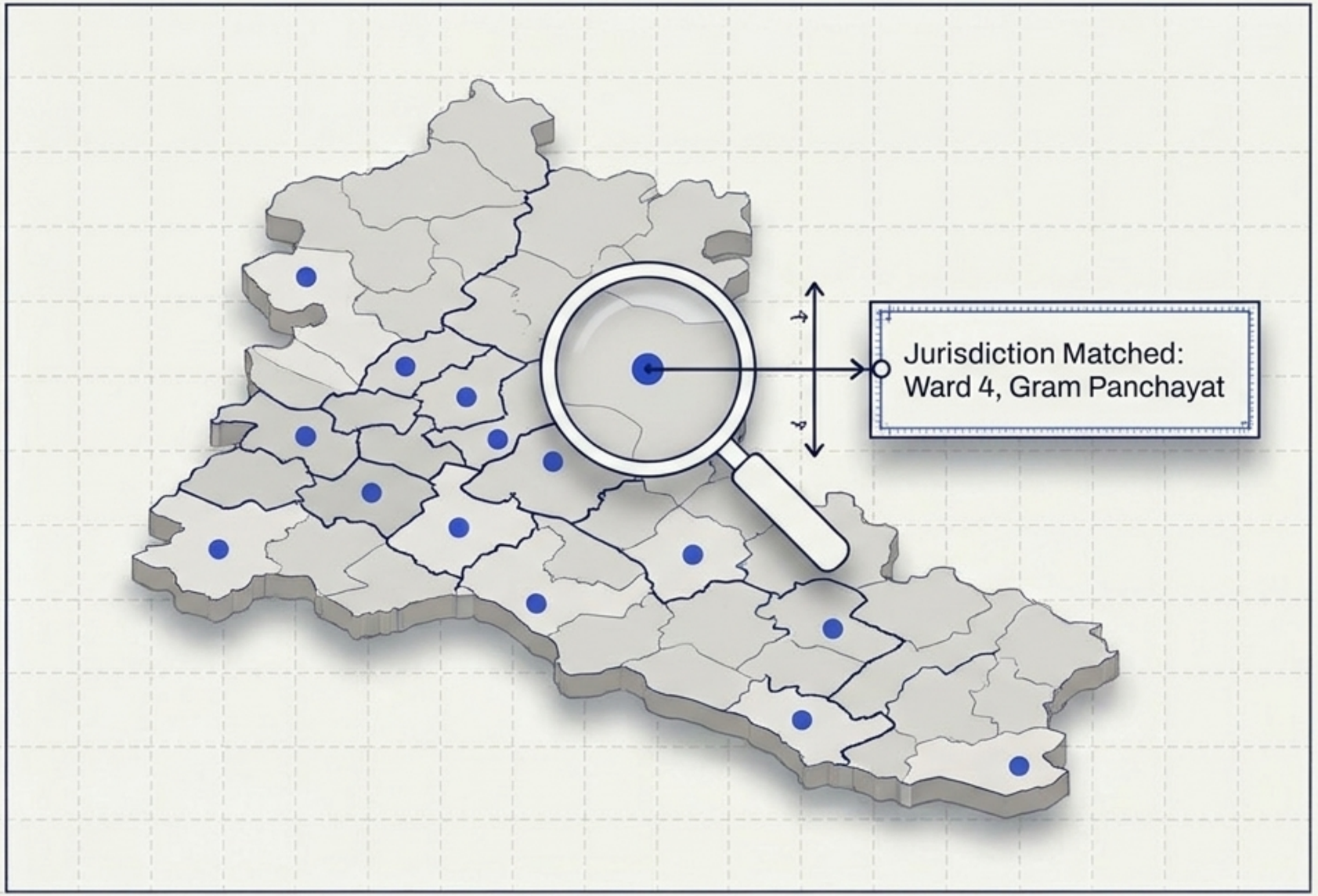
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Geographic Intelligence and Spatial Mapping

PostGIS Spatial Database

Administrative Boundary Polygons

Real-time Coordinate Translation



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Automated Hierarchical Routing Logic

Rules engine eliminates manual dispatching errors, ensuring the exact local authority receives the ticket instantly based on spatial boundaries and issue category.

Trigger: Verified AI Event + GIS Coordinate

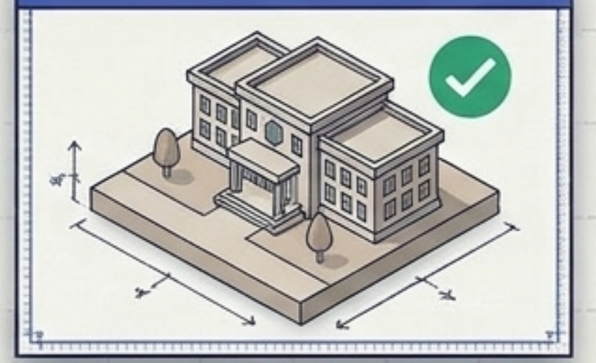
Issue = Village Road / Local Sanitation

Route to Gram Panchayat



Issue = Block-level Infrastructure

Route to Taluk Panchayat



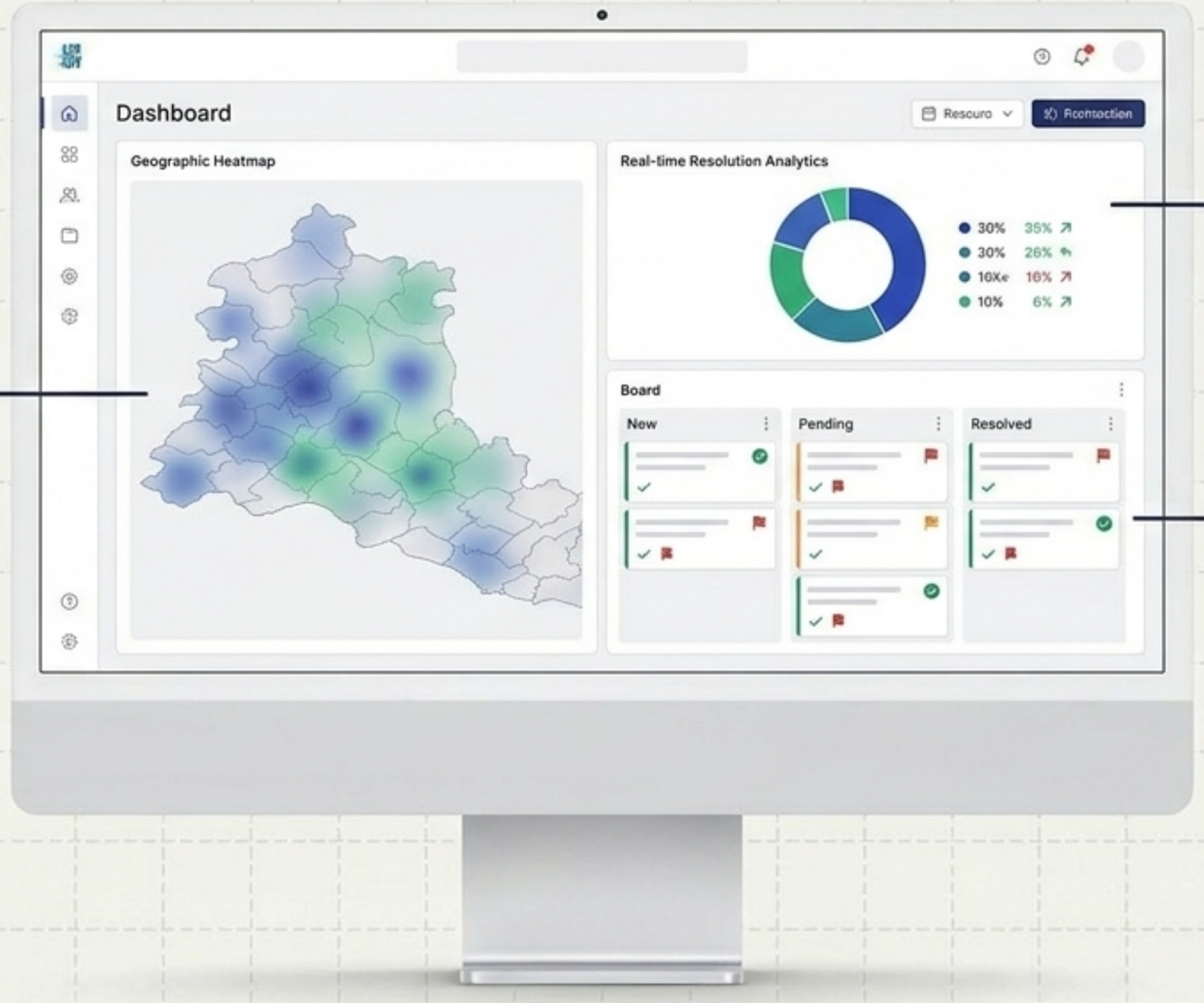
Issue = District Highway / Major Utility

Route to Zilla Panchayat



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The Official Command Dashboard



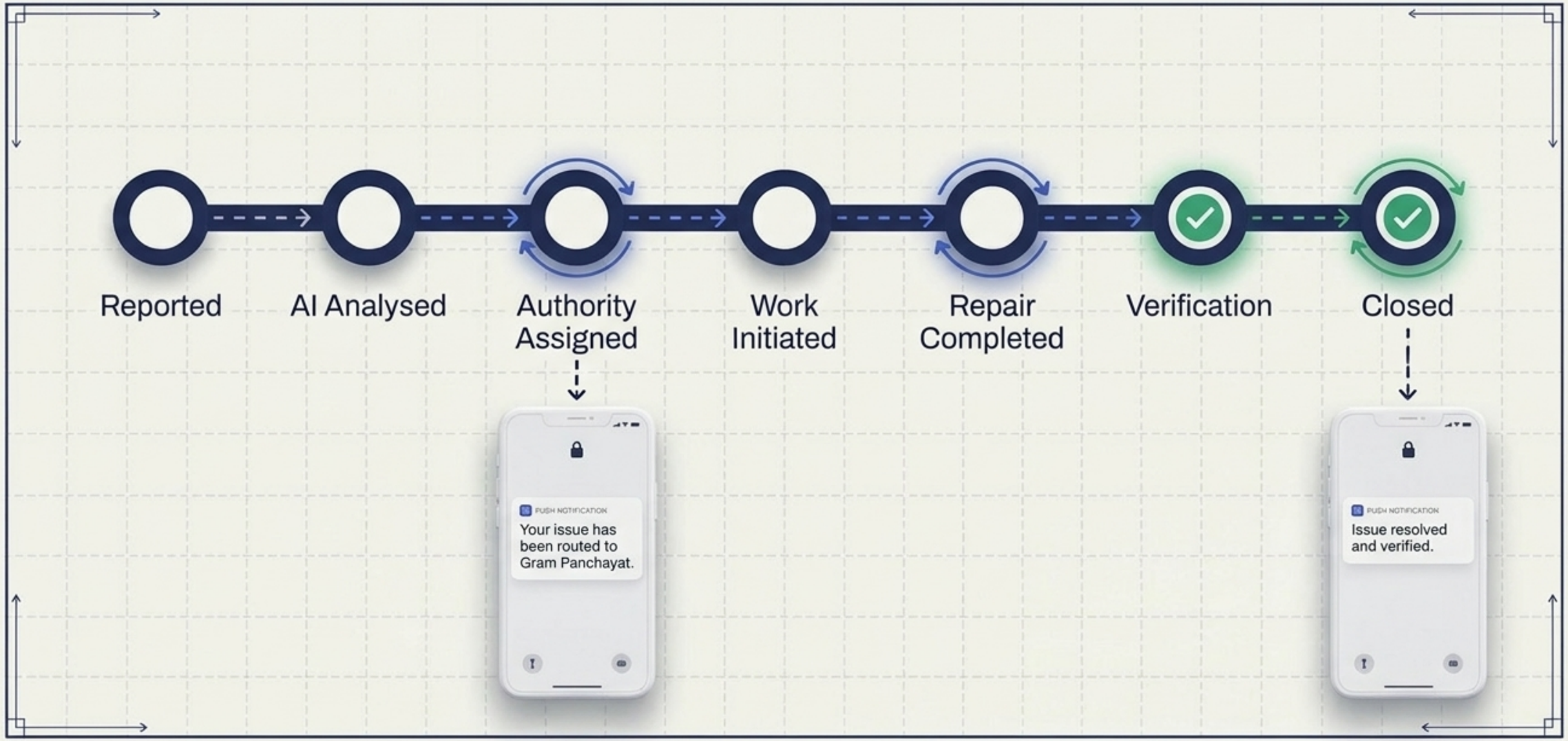
Geographic Resource Allocation Heatmaps

Real-time Resolution Analytics

Automated Priority Alerts

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End-to-End Issue Resolution Workflow



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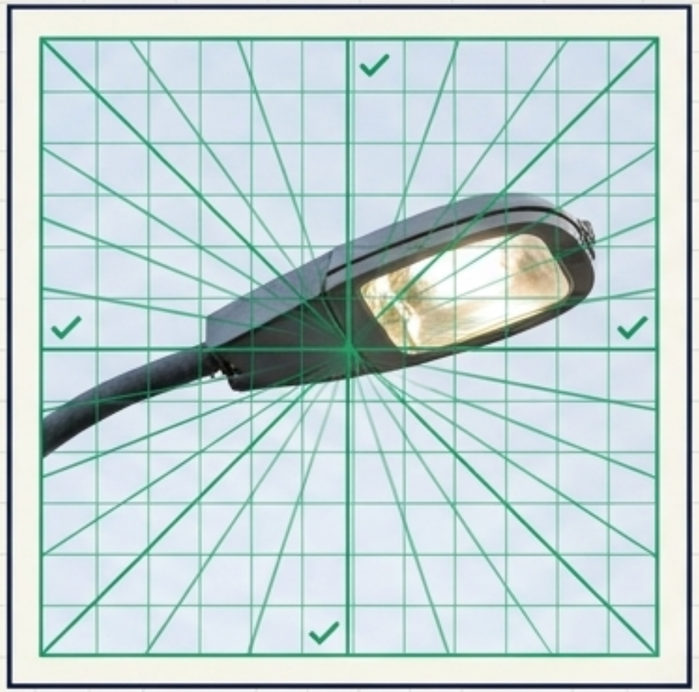
Proof of Work: AI Repair Verification

Machine learning compares baseline images against resolution claims to ensure infrastructure has actually been repaired, preventing false closures.



Baseline Signature Captured

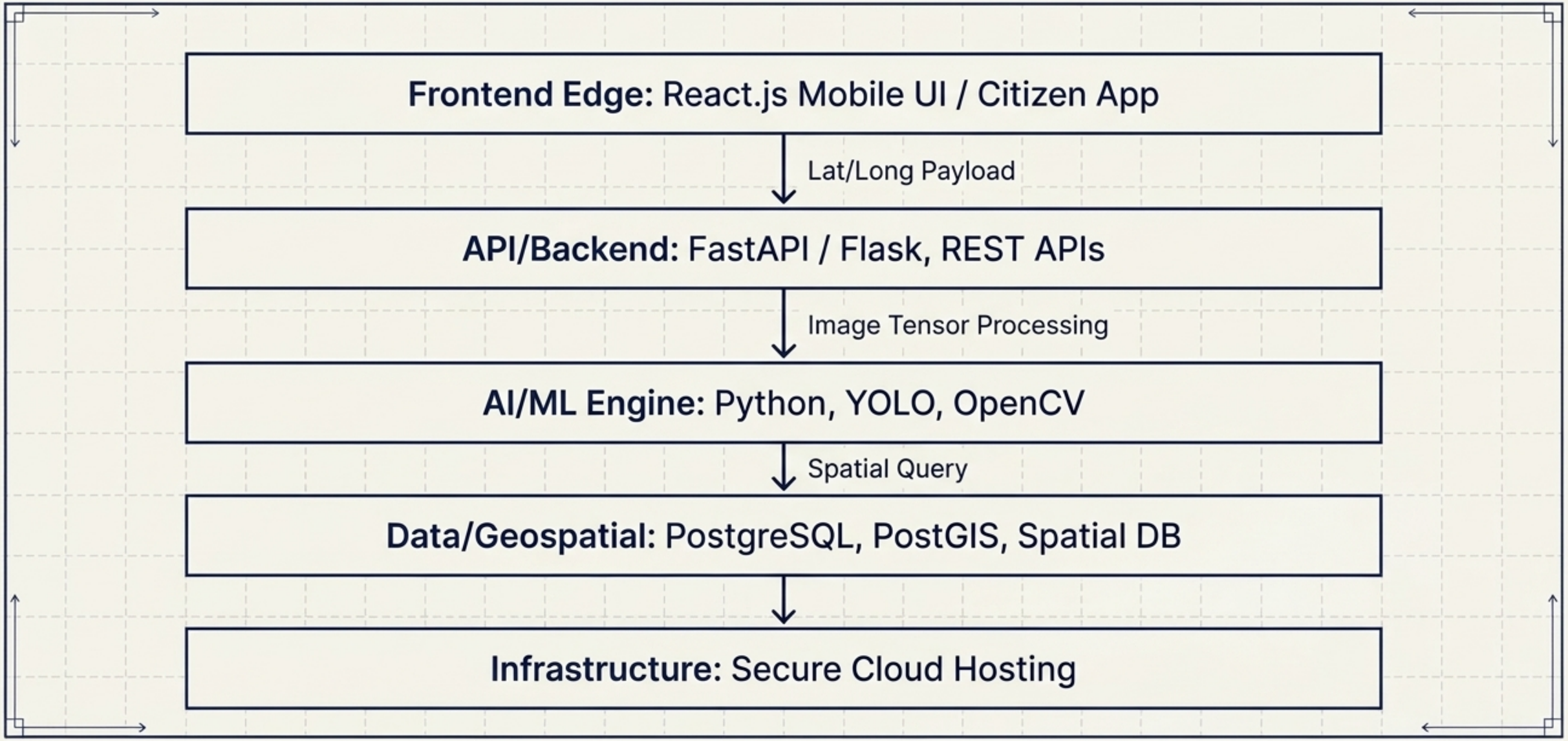




Resolution Signature Verified

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Comprehensive System Architecture



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Technology Stack & Training Datasets



React



HTML/CSS/JS



Python



FastAPI



YOLO



OpenCV



PostgreSQL



PostGIS



Maps API

Model Training

Leveraging open-source civic infrastructure datasets (e.g., global road damage datasets) combined with custom annotated imagery to train highly localized classification models.

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Security, Privacy & Data Governance





Authentication

Secure login and strict Role-Based Access Control (RBAC) for government officials.



Data Privacy

End-to-end anonymization of citizen data and secure cloud environments.

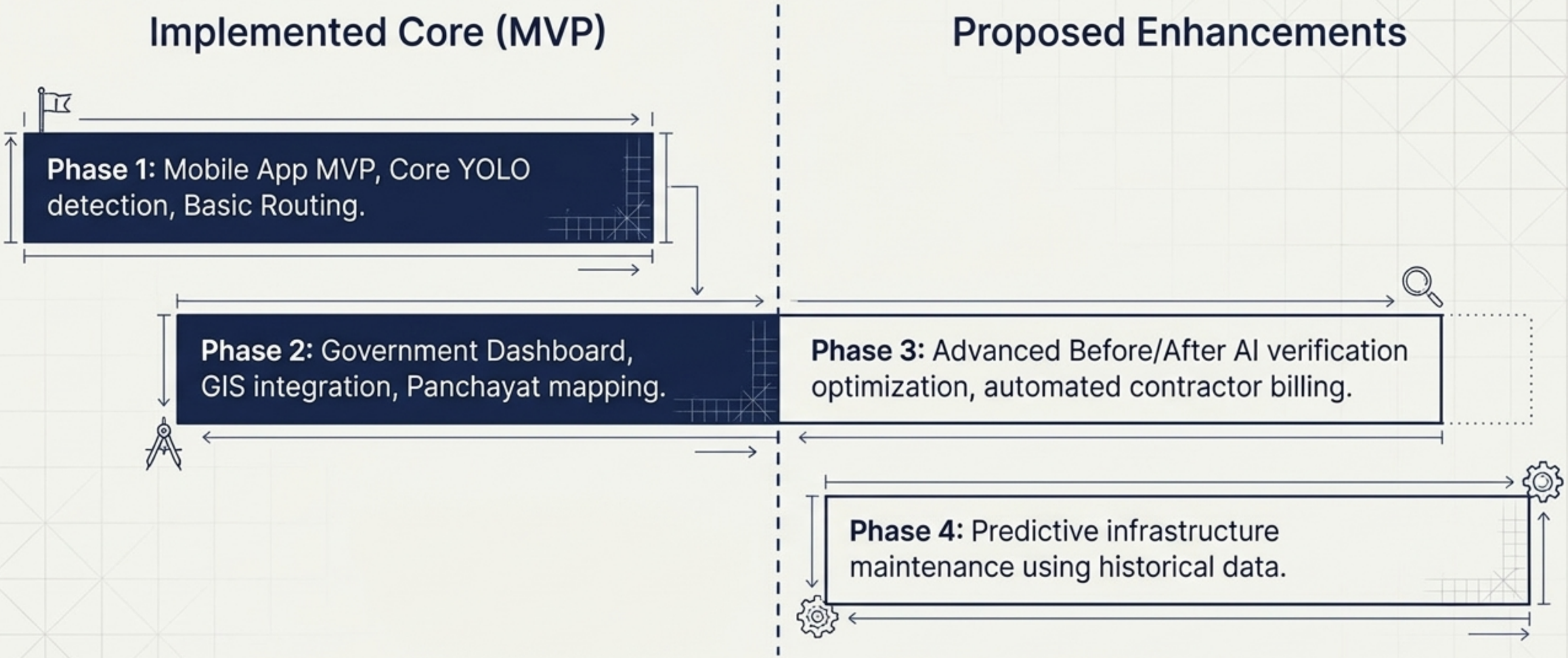


System Integrity

GPS spoofing prevention. Reports require real-time, live camera capture (gallery uploads disabled) to guarantee event authenticity.

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Implementation Roadmap

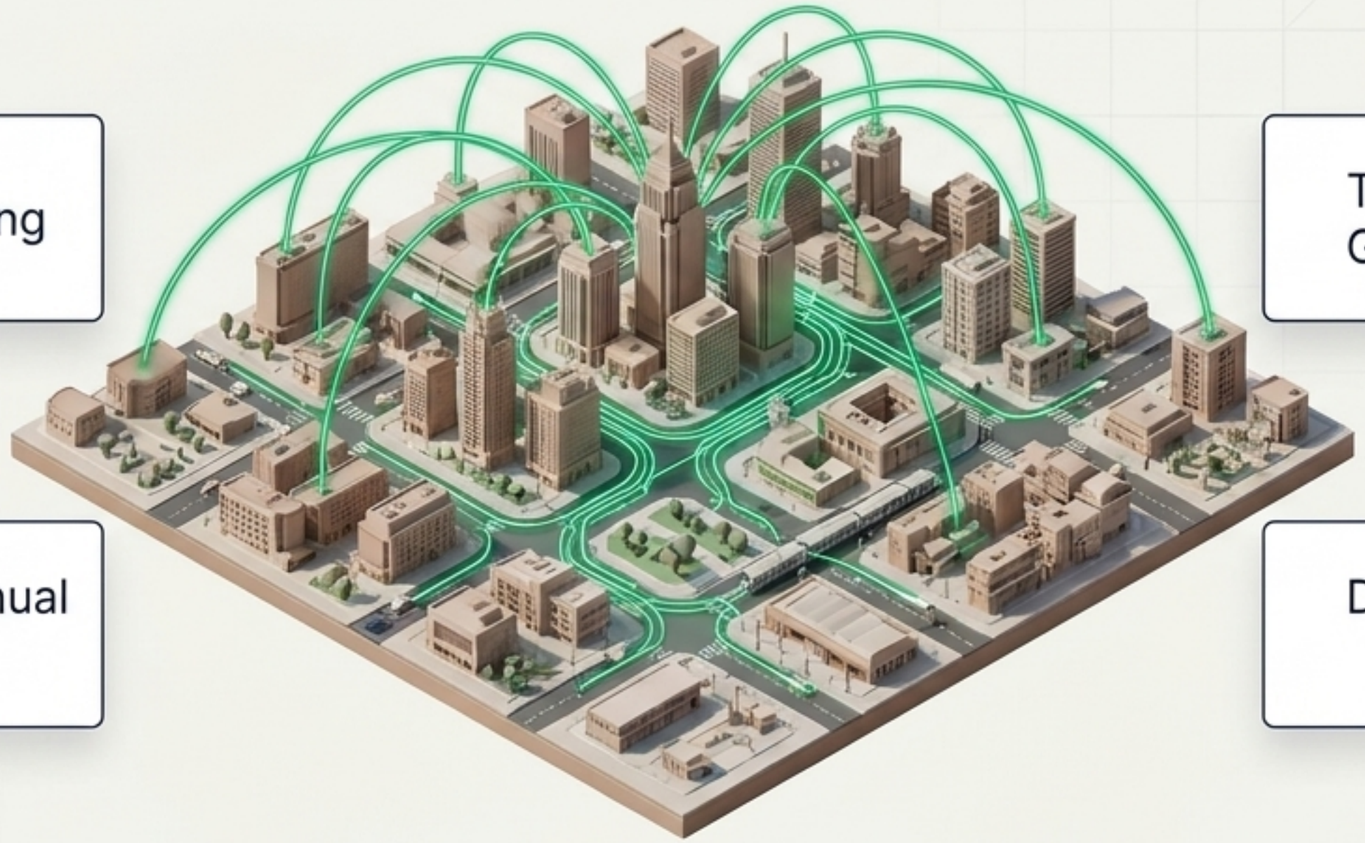


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Transforming Civic Ecosystems

Faster Routing

Reduced Manual Overhead



Transparent Governance

Data-Driven Planning

NEXUS AI connects citizens, AI, and local authorities to build a smarter, accountable civic ecosystem.

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